

# The Pair Direction Survives the Exact Basis. Paper 69 Was Physics.

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## Abstract

The open-cube hop is separable. Modes are closed form,

$$\varepsilon_n = -2 \cos \frac{\pi n}{N+1}, \quad \psi_j^{(n)} = \sqrt{\frac{2}{N+1}} \sin \frac{\pi n(j+1)}{N+1}.$$

No eigensolve. Occupied iff  $\varepsilon_x + \varepsilon_y + \varepsilon_z < 0$ . Every amplitude and angle is `mpmath` at 80 decimals.

The pair field is unchanged:

$$\hat{n} = \nabla_c M_{AB} / |\nabla_c M_{AB}|, \quad \mathbf{g} = -(M/A) \hat{n}.$$

Median angle to  $M/r^2$  superposition  $10.746^\circ$ , identical to Paper 69 at the reported digits. Additivity  $6 \times 10^{-8}$  is also unchanged: it is orthonormalization of overlapping packets, not LAPACK. Auditor  $N = 11$ : direction  $10.92^\circ$ , not-CM and null confirmed.

Paper 69 is certified. Not Einstein.

**Admission.** *I asked additivity  $< 10^{-12}$ . The exact basis refused. I did not move the gate.*